

Materials Engineering Reflection

During this module of the FYRE program, we were tasked with creating a sensor that would be able to detect differences in humidity. Along with this main goal, we were given information about photolithography, circuits, and we were given tours of the clean room and Wilbur Powerhouse.

In my experience, the process of creating a sensor was taken with an extremely creative and open-minded approach. None of the professors ever threw an idea completely out of the equation or ever labeled anything as “bad”. They took opposing ideas openly, and we eventually were led to the idea of creating an electrode. We created a map for the electrode in Wilbur Powerhouse and used it as a sort of stencil to lay down conductive material. We made electrodes out of Silver and Carbon. We then measured the resistivity in order to learn more about the material. After that, we added a polymer called Poly Ethylene Dioxythiophene Poly Styrene Suffonate, or PEDOT-PSS for short. PEDOT-PSS is a polymer that is conductive, but has bonds that can be disrupted when in the presence of water, making it less conductive. This quality allows for the electrode to act as a sensor, because it will detect water and then the resistivity will change. We tested this, and it worked perfectly. The resistivity of the electrode was significantly altered after we breathed on the electrode due to the rapid change of humidity breaking apart the bonds of the PEDOT-PSS.

We were also given lessons in photolithography and spin-coating while in the clean room. They discussed how it is important to keep dust particles out of the clean room due to the size and precision that it takes to spin-coat and pattern material. Since the process involves using literal photons, one particle of dust can ruin the patterning process. Additionally, during our own module, we were given exposure to spin coating when we put the PEDOT-PSS onto the electrodes. There was definitely room for experimental error throughout the entire process, but we had one electrode that worked solidly. An important thing that I learned was that it is important to have multiple trials in order to try to negate the effects of experimental error. There is a lot that can go wrong during the creation or testing of anything, so it is important to keep that in mind.

At the end of the week, we were given exposure to circuit design. Very basic, but we created a circuit on paper with four lights in parallel which was fueled by a lithium battery. Throughout all of the modules, we were given the ability to fail, which I feel is very beneficial to my learning and improvement. The FYRE module so far has been helpful and informative to my development. The thing that I am most grateful for is that the entire process has been heavily guided through curiosity and experimentation.